

# Package ‘ETAINDoseCalculator’

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**Type** Package

**Title** Functions for RF-EMF Dose Calculations

**Version** 0.2.0

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**Description** Package for calculating RF-  
EMF doses for cohorts. Calculations are based on the ETAIN Dose model.

**License** GPL-3

**Encoding** UTF-8

**LazyData** true

**RoxygenNote** 7.3.2

**Roxygen** list(markdown = TRUE)

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tidyr,  
yaml

**Suggests** knitr,  
rmarkdown,  
testthat (>= 3.0.0)

**VignetteBuilder** knitr

**Depends** R (>= 3.5)

**Config/testthat/edition** 3

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|--------------|-----------|

---

|                                 |                                                   |
|---------------------------------|---------------------------------------------------|
| calculate_data_tech_proportions | <i>Get mobile data technology use proportions</i> |
|---------------------------------|---------------------------------------------------|

---

## Description

Based on user variable use\_5g that indicates if participant uses 5G on mobile phone or not.

## Usage

```
calculate_data_tech_proportions(use_5g, params)
```

## Arguments

|        |                                                                 |
|--------|-----------------------------------------------------------------|
| use_5g | boolean variable (TRUE if participant uses 5G, FALSE otherwise) |
| params | parameter list                                                  |

## Value

list with technology use proportions

---

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| calculate_emf_doses | <i>Calculate Total RF-EMF Doses for ALL samples</i> |
|---------------------|-----------------------------------------------------|

---

### Description

Calculate Total RF-EMF Doses for ALL samples

### Usage

```
calculate_emf_doses(data, param_file = NULL, default_value_file = NULL)
```

### Arguments

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data               | A data frame. Must have the following columns: sample_id: unique participant identifier mpc_duration: daily duration of mobile phone call in seconds mpc_ear_prop: proportion of time phone is held against ear during call urbanicity: urban, suburban, or rural mpd_duration: daily duration of mobile data traffic in seconds mpd_wifi_prop: proportion of time WiFi is used (vs data) for mobile data traffic dect_duration: daily duration of DECT/cordless phone call in seconds dect_ear_prop: lptp_duration: tblt_duration: wifi_duration: |
| param_file         | optional path to YAML parameter configuration file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| default_value_file | optional path to YAML default value file in case of missing data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Value

A data frame. Columns named SOURCE\_dose\_TISSUE contain RF-EMF dose of each participant in mJ/kg/day

---

|                                |                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------|
| calculate_location_proportions | <i>Calculate proportion of time spent at home vs work vs outdoors vs travelling</i> |
|--------------------------------|-------------------------------------------------------------------------------------|

---

### Description

Based on travel time (input variable by user), assuming a fixed ration for time spent at home vs work vs outdoors

### Usage

```
calculate_location_proportions(travel_time, home_prop, work_prop, outd_prop)
```

### Arguments

|             |                                                              |
|-------------|--------------------------------------------------------------|
| travel_time | daily time spent travelling / commuting in seconds (s)       |
| home_prop   | proportion of time spent at home WITHOUT considering commute |
| work_prop   | proportion of time spent at work WITHOUT considering commute |
| outd_prop   | proportion of time spent outside WITHOUT considering commute |

**Value**

list with proportions

---

|                |                                      |
|----------------|--------------------------------------|
| check_duration | <i>Check input values - duration</i> |
|----------------|--------------------------------------|

---

**Description**

Check input values - duration

**Usage**

check\_duration(duration)

**Arguments**

duration                  duration

**Value**

TRUE if duration is valid, FALSE if duration is not valid

---

|                        |                                   |
|------------------------|-----------------------------------|
| check_input_param_list | <i>Check input parameter list</i> |
|------------------------|-----------------------------------|

---

**Description**

Check input parameter list

**Usage**

check\_input\_param\_list(params)

**Arguments**

params                  parameter list in YAML format

**Value**

TRUE if parameter list is valid, FALSE if parameter list is invalid

---

|                   |                                         |
|-------------------|-----------------------------------------|
| check_proportions | <i>Check input values - proportions</i> |
|-------------------|-----------------------------------------|

---

**Description**

Check input values - proportions

**Usage**

```
check_proportions(proportions)
```

**Arguments**

proportions      Single proportion avlue or vector with proportions to check

**Value**

TRUE if proportions are valid, FALSE if invalid

---

|              |                                 |
|--------------|---------------------------------|
| example_data | <i>Example data for testing</i> |
|--------------|---------------------------------|

---

**Description**

A dataset containing example user data (made up)

**Usage**

```
example_data
```

**Format**

A data frame with 50 rows

**dect\_duration** test

**dect\_ear\_prop** test

**gaming\_duration** test

**headp\_ear\_num** test

**headphone\_duration** test

**hotspot\_duration** test

**lptp\_duration** test

**mpc\_duration** test

**mpc\_ear\_prop** test

**mpc\_headp\_prop** test

**mpd\_duration** test

**mpd\_high\_dt\_dur** test

**mpd\_high\_dt\_prop** test  
**mpd\_low\_dt\_dur** test  
**mpd\_lowmed\_dt\_dur** test  
**mpd\_medhigh\_dt\_dur** test  
**mpd\_wifi\_prop** test  
**sample\_id** test  
**smarthome\_duration** test  
**smartwatch\_duration** test  
**tblt\_duration** test  
**tracker\_duration** test  
**travel\_time** test  
**urbanicity** test  
**use\_5g** test  
**vr\_duration** test  
**wifi\_duration** test

### Source

Generated for package testing.

---

fill\_missing\_variables

*Fill missing variables*

---

### Description

Fill missing variables

### Usage

```
fill_missing_variables(data, defaults, warn_threshold = 0.1)
```

### Arguments

|                |                                                       |
|----------------|-------------------------------------------------------|
| data           | data                                                  |
| defaults       | defaults                                              |
| warn_threshold | threshold proportion of missing data to raise warning |

### Value

data frame with missing data replaced with default values

---

|                       |                                                              |
|-----------------------|--------------------------------------------------------------|
| flat_params_to_nested | <i>Convert flat parameter table to nested parameter YAML</i> |
|-----------------------|--------------------------------------------------------------|

---

**Description**

Convert flat parameter table to nested parameter YAML

**Usage**

```
flat_params_to_nested(flat)
```

**Arguments**

|      |                                                           |
|------|-----------------------------------------------------------|
| flat | Flat parameter data frame with "path" and "value" columns |
|------|-----------------------------------------------------------|

**Value**

nested parameter list

---

|                   |                                                               |
|-------------------|---------------------------------------------------------------|
| get_act_pwr_props | <i>Get low/lowmed/medhigh/high activity power proportions</i> |
|-------------------|---------------------------------------------------------------|

---

**Description**

Get low/lowmed/medhigh/high activity power proportions

**Usage**

```
get_act_pwr_props(low_dur, lowmed_dur, medhigh_dur, high_dur)
```

**Arguments**

|             |     |
|-------------|-----|
| low_dur     | ... |
| lowmed_dur  | ... |
| medhigh_dur | ... |
| high_dur    | ... |



---

|                   |                                             |
|-------------------|---------------------------------------------|
| get_cordless_dose | <i>Calculate Dose from Cordless Calling</i> |
|-------------------|---------------------------------------------|

---

**Description**

Calculate Dose from Cordless Calling

**Usage**

```
get_cordless_dose(duration, ear_proportion, params)
```

**Arguments**

|                |                                                  |
|----------------|--------------------------------------------------|
| duration       | Duration of cordless calls in seconds            |
| ear_proportion | Proportion of holding cordless phone against ear |
| params         | Parameter list                                   |

**Value**

List with brain dose and body dose in mJ/kg/day

---

|                  |                                                     |
|------------------|-----------------------------------------------------|
| get_cordless_pwr | <i>Calculate Aggregated Power of Cordless Phone</i> |
|------------------|-----------------------------------------------------|

---

**Description**

Calculate Aggregated Power of Cordless Phone

**Usage**

```
get_cordless_pwr(params)
```

**Arguments**

|        |       |
|--------|-------|
| params | descr |
|--------|-------|

**Value**

descr

---

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| get_cordless_sar | <i>Calculate Aggregated SAR during cordless call (tissue-specific)</i> |
|------------------|------------------------------------------------------------------------|

---

**Description**

Calculate Aggregated SAR during cordless call (tissue-specific)

**Usage**

```
get_cordless_sar(ear_proportion, params, tissue_params)
```

**Arguments**

|                |                                                  |
|----------------|--------------------------------------------------|
| ear_proportion | Proportion of holding cordless phone against ear |
| params         | descr                                            |
| tissue_params  | descr                                            |

**Value**

sar

---

|                   |                                               |
|-------------------|-----------------------------------------------|
| get_farfield_dose | <i>Calculate Dose from far-field exposure</i> |
|-------------------|-----------------------------------------------|

---

**Description**

Calculate Dose from far-field exposure

**Usage**

```
get_farfield_dose(urbanicity, travel_time, params)
```

**Arguments**

|             |                                  |
|-------------|----------------------------------|
| urbanicity  | Urbanicity of home environment   |
| travel_time | Time per day spent commuting (s) |
| params      | Parameter list                   |

**Value**

List with brain dose and body dose in mJ/kg/day

---

|                  |                                                      |
|------------------|------------------------------------------------------|
| get_farfield_pwr | <i>Calculate far-field exposure aggregated power</i> |
|------------------|------------------------------------------------------|

---

**Description**

Calculate far-field exposure aggregated power

**Usage**

```
get_farfield_pwr(urb_list, travel_time, params)
```

**Arguments**

|          |                                                                          |
|----------|--------------------------------------------------------------------------|
| urb_list | list of binary urbanicity values created with recode_urbanicity function |
| params   | description                                                              |

**Value**

far-field power

---

|                  |                                         |
|------------------|-----------------------------------------|
| get_farfield_sar | <i>Calculate far-field exposure SAR</i> |
|------------------|-----------------------------------------|

---

**Description**

This function will be expanded in the future to provide a more fine-tuned calculation of the far-field SAR values.

**Usage**

```
get_farfield_sar(params, tissue_params)
```

**Arguments**

|               |       |
|---------------|-------|
| params        | descr |
| tissue_params | descr |

**Value**

aggregated sar

---

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| get_gaming_dose | <i>Calculate brain and body dose from gaming with a portable console</i> |
|-----------------|--------------------------------------------------------------------------|

---

**Description**

Calculate brain and body dose from gaming with a portable console

**Usage**

```
get_gaming_dose(duration_gaming, params)
```

**Arguments**

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| duration_gaming | Daily gaming duration with portable console in seconds |
| params          | Parameter list                                         |

**Value**

list with daily brain and body dose from smart home

---

|                    |                                                                      |
|--------------------|----------------------------------------------------------------------|
| get_headphone_dose | <i>Calculate brain and body dose from using bluetooth headphones</i> |
|--------------------|----------------------------------------------------------------------|

---

**Description**

Important: exposure comes from both mobile phone and from smart watch.

**Usage**

```
get_headphone_dose(duration_headphones, params)
```

**Arguments**

|                     |                                                          |
|---------------------|----------------------------------------------------------|
| duration_headphones | Daily duration of using bt headphones connected to phone |
| params              | Parameter list                                           |

**Value**

list with daily brain and body dose from bt headphones connected to phone

---

|                      |                                          |
|----------------------|------------------------------------------|
| get_headp_headp_dose | <i>Calculate contribution from headp</i> |
|----------------------|------------------------------------------|

---

**Description**

Calculate contribution from headp

**Usage**

```
get_headp_headp_dose(duration_headphones, params, tissue_params)
```

**Arguments**

|                     |                                            |
|---------------------|--------------------------------------------|
| duration_headphones | Daily duration of headphone use in seconds |
| params              | Parameter list                             |
| tissue_params       | tissue-specific parameter list             |

---

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| get_headp_phone_dose | <i>Calculate contribution from mobile phone while using headp</i> |
|----------------------|-------------------------------------------------------------------|

---

**Description**

Calculate contribution from mobile phone while using headp

**Usage**

```
get_headp_phone_dose(duration_headphones, params, tissue_params)
```

**Arguments**

|                     |                                            |
|---------------------|--------------------------------------------|
| duration_headphones | Daily duration of headphone use in seconds |
| params              | Parameter list                             |
| tissue_params       | tissue-specific parameter list             |

---

|                     |                                                                            |
|---------------------|----------------------------------------------------------------------------|
| get_headp_phone_pwr | <i>Calculate power from bluetooth headphones (phone contribution only)</i> |
|---------------------|----------------------------------------------------------------------------|

---

**Description**

Calculate power from bluetooth headphones (phone contribution only)

**Usage**

get\_headp\_phone\_pwr(params)

**Arguments**

params                  Parameter list

**Value**

output power

---

|                     |                                                                          |
|---------------------|--------------------------------------------------------------------------|
| get_headp_phone_sar | <i>Calculate SAR from bluetooth headphones (phone contribution only)</i> |
|---------------------|--------------------------------------------------------------------------|

---

**Description**

Calculate SAR from bluetooth headphones (phone contribution only)

**Usage**

get\_headp\_phone\_sar(params, tissue\_params)

**Arguments**

params                  Parameter list  
tissue\_params          Tissue-specific parameter list (SAR values)

**Value**

SAR value

---

|               |                                                                                |
|---------------|--------------------------------------------------------------------------------|
| get_headp_pwr | <i>Calculate power from bluetooth headphones (headphone contribution only)</i> |
|---------------|--------------------------------------------------------------------------------|

---

**Description**

Calculate power from bluetooth headphones (headphone contribution only)

**Usage**

```
get_headp_pwr(params)
```

**Arguments**

|        |                |
|--------|----------------|
| params | Parameter list |
|--------|----------------|

**Value**

output power

---

|               |                                                                          |
|---------------|--------------------------------------------------------------------------|
| get_headp_sar | <i>Calculate SAR from bluetooth headphones (watch contribution only)</i> |
|---------------|--------------------------------------------------------------------------|

---

**Description**

Calculate SAR from bluetooth headphones (watch contribution only)

**Usage**

```
get_headp_sar(params, tissue_params)
```

**Arguments**

|               |                                             |
|---------------|---------------------------------------------|
| params        | Parameter list                              |
| tissue_params | Tissue-specific parameter list (SAR values) |

**Value**

SAR value

---

|                  |                                                                         |
|------------------|-------------------------------------------------------------------------|
| get_hotspot_dose | <i>Calculate brain and body dose from using mobile phone as hotspot</i> |
|------------------|-------------------------------------------------------------------------|

---

**Description**

Calculate brain and body dose from using mobile phone as hotspot

**Usage**

```
get_hotspot_dose(duration_hotspot, params)
```

**Arguments**

|                  |                                                     |
|------------------|-----------------------------------------------------|
| duration_hotspot | Daily duration of using phone as hotspot in seconds |
| params           | Parameter list                                      |

**Value**

list with daily brain and body dose from hotspot use

---

|                 |                                                                  |
|-----------------|------------------------------------------------------------------|
| get_hotspot_pwr | <i>Calculate output power from using mobile phone as hotspot</i> |
|-----------------|------------------------------------------------------------------|

---

**Description**

Calculate output power from using mobile phone as hotspot

**Usage**

```
get_hotspot_pwr(params)
```

**Arguments**

|        |                |
|--------|----------------|
| params | Parameter list |
|--------|----------------|

**Value**

Aggregated output power of mobile phone hotspot in mW



---

|                 |                                   |
|-----------------|-----------------------------------|
| get_hotspot_sar | <i>Calculate SAR from hotspot</i> |
|-----------------|-----------------------------------|

---

**Description**

Calculate SAR from hotspot

**Usage**

```
get_hotspot_sar(params, tissue_params)
```

**Arguments**

|               |                |
|---------------|----------------|
| params        | Parameter list |
| tissue_params | Tissue params  |

**Value**

tissue-specific SAR from hotspot

---

|                 |                                       |
|-----------------|---------------------------------------|
| get_laptop_dose | <i>Calculate Dose from Laptop Use</i> |
|-----------------|---------------------------------------|

---

**Description**

Calculate Dose from Laptop Use

**Usage**

```
get_laptop_dose(duration, params)
```

**Arguments**

|          |                                   |
|----------|-----------------------------------|
| duration | Duration of laptop use in seconds |
| params   | Parameter list                    |

**Value**

List with brain dose and body dose in mJ/kg/day

---

|                |                                          |
|----------------|------------------------------------------|
| get_laptop_pwr | <i>Calculate Laptop Aggregated Power</i> |
|----------------|------------------------------------------|

---

**Description**

Calculate Laptop Aggregated Power

**Usage**

get\_laptop\_pwr(params)

**Arguments**

|        |      |
|--------|------|
| params | text |
|--------|------|

**Value**

text

---

|                |                                        |
|----------------|----------------------------------------|
| get_laptop_sar | <i>Calculate Laptop Aggregated SAR</i> |
|----------------|----------------------------------------|

---

**Description**

Calculate Laptop Aggregated SAR

**Usage**

get\_laptop\_sar(params, tissue\_params)

**Arguments**

|               |       |
|---------------|-------|
| params        | descr |
| tissue_params | descr |

**Value**

Aggregated tissue SAR from laptop use (W/kg/W)

---

get\_mobilecall\_bt\_headp\_dose

*Calculate RF-EMF dose from mobile calling using Bluetooth headphones (headphone contribution)*

---

## Description

Calculates the RF-EMF dose from mobile calling using Bluetooth headphones (headphone contribution).

## Usage

```
get_mobilecall_bt_headp_dose(
    duration,
    headp_prop,
    headp_ear_num,
    params,
    tissue_params
)
```

## Arguments

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| duration      | Duration of mobile phone call in seconds per day                    |
| headp_prop    | Proportion of time Bluetooth headphones are used during mobile call |
| headp_ear_num | If participant uses 1 or 2 Bluetooth headphones during mobile call  |
| params        | Parameter list                                                      |
| tissue_params | Tissue parameter list                                               |

## Details

The dose is calculated as:

$$Dose = Duration \times Power \times SAR \times headp_{prop} \times headp_{ear\_num}$$

Where:

- *Duration* is the duration of mobile phone call in seconds per day
- *SAR* is the specific absorption rate from Bluetooth in W/kg/W
- *Power* is the Bluetooth output power of the mobile phone during mobile calling mW
- *headp<sub>prop</sub>* is the proportion of time Bluetooth headphones are used during mobile calls
- *headp<sub>ear<sub>num</sub></sub>* is the number of Bluetooth headphones (1 or 2) used during mobile calls

## Value

Dose from mobile calling using Bluetooth headphones in mJ/kg/day (headphone contribution)

## See Also

[get\\_mobilecall\\_bt\\_phone\\_dose\(\)](#), [get\\_mobilecall\\_bt\\_headp\\_pwr\(\)](#), [get\\_mobilecall\\_bt\\_headp\\_sar\(\)](#)

---

```
get_mobilecall_bt_headp_pwr
```

*Calculate output power from mobile calling using Bluetooth headphones (headphone contribution)*

---

### Description

Calculates the output power from mobile calling using Bluetooth headphones (headphone Bluetooth contribution).

### Usage

```
get_mobilecall_bt_headp_pwr(params)
```

### Arguments

|        |                |
|--------|----------------|
| params | Parameter list |
|--------|----------------|

### Details

In this version, this function serves as a placeholder and simply fetches the corresponding parameter from the parameter list.

### Value

Output power from mobile phone (Bluetooth contribution) during mobile phone call in mW

---

```
get_mobilecall_bt_headp_sar
```

*Calculate SAR from mobile calling using Bluetooth headphones (headphone contribution)*

---

### Description

Calculates SAR from mobile calling using Bluetooth headphones (headphone Bluetooth contribution).

### Usage

```
get_mobilecall_bt_headp_sar(params, tissue_params)
```

### Arguments

|               |                       |
|---------------|-----------------------|
| params        | Parameter list        |
| tissue_params | Tissue parameter list |

### Details

In this version, this function serves as a placeholder and simply fetches the corresponding parameter from the parameter list.

**Value**

Output power from mobile phone (Bluetooth contribution) during mobile phone call in mW

---

get\_mobilecall\_bt\_phone\_dose

*Calculate RF-EMF dose from mobile calling using Bluetooth headphones (phone contribution)*

---

**Description**

Calculates the RF-EMF dose from mobile calling using Bluetooth headphones (phone contribution).

**Usage**

get\_mobilecall\_bt\_phone\_dose(duration, headp\_prop, params, tissue\_params)

**Arguments**

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| duration      | Duration of mobile phone call in seconds per day                    |
| headp_prop    | Proportion of time Bluetooth headphones are used during mobile call |
| params        | Parameter list                                                      |
| tissue_params | Tissue parameter list                                               |

**Details**

The dose is calculated as:

$$Dose = Duration \times Power \times SAR$$

Where:

- *Duration* is the duration of mobile phone call in seconds per day
- *SAR* is the specific absorption rate from Bluetooth in W/kg/W
- *Power* is the Bluetooth output power of the mobile phone during mobile calling mW

**Value**

Dose from mobile calling using Bluetooth headphones in mJ/kg/day (phone contribution)

**See Also**

[get\\_mobilecall\\_bt\\_headp\\_dose\(\)](#), [get\\_mobilecall\\_bt\\_phone\\_pwr\(\)](#), [get\\_mobilecall\\_bt\\_phone\\_sar\(\)](#)

---

```
get_mobilecall_bt_phone_pwr
```

*Calculate output power from mobile calling using Bluetooth headphones (phone contribution)*

---

### Description

Calculates the output power from mobile calling using Bluetooth headphones (phone Bluetooth contribution).

### Usage

```
get_mobilecall_bt_phone_pwr(params)
```

### Arguments

|        |                |
|--------|----------------|
| params | Parameter list |
|--------|----------------|

### Details

In this version, this function serves as a placeholder and simply fetches the corresponding parameter from the parameter list.

### Value

Output power from mobile phone (Bluetooth contribution) during mobile phone call in mW

---

```
get_mobilecall_bt_phone_sar
```

*Calculate SAR from mobile calling using Bluetooth headphones (phone contribution)*

---

### Description

Calculates the specific absorption rate from mobile calling using Bluetooth headphones (phone contribution).

### Usage

```
get_mobilecall_bt_phone_sar(params, tissue_params)
```

### Arguments

|               |                       |
|---------------|-----------------------|
| params        | Parameter list        |
| tissue_params | Tissue parameter list |

### Value

SAR from mobile calling using Bluetooth headphones in W/kg/W (phone contribution)

---

|                     |                                                                               |
|---------------------|-------------------------------------------------------------------------------|
| get_mobilecall_dose | <i>Calculate RF-EMF dose from mobile calling (mobile phone and Bluetooth)</i> |
|---------------------|-------------------------------------------------------------------------------|

---

### Description

Includes all sources: RF-EMF from phone itself, for Bluetooth connection to headphones, and from headphones themselves

### Usage

```
get_mobilecall_dose(
    duration,
    ear_prop,
    headp_prop,
    urbanicity,
    use_5g,
    travel_time,
    headp_ear_num,
    params
)
```

### Arguments

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| duration      | Duration of mobile phone call in seconds per day                                                        |
| ear_prop      | Proportion of time mobile phone is held against ear during call                                         |
| headp_prop    | Proportion of time Bluetooth headphones are used during mobile call while phone is NOT held against ear |
| urbanicity    | Urbanicity of home / workplace                                                                          |
| use_5g        | TRUE if participant uses 5G services on mobile phone, FALSE if not                                      |
| travel_time   | Time spent commuting in seconds per day                                                                 |
| headp_ear_num | If participant uses 1 or 2 Bluetooth headphones during mobile call                                      |
| params        | Parameter list                                                                                          |

### Details

The mobile call RF-EMF dose is calculated as:

$$Dose_{mobilecall} = Dose_{phone} + Dose_{phoneBluetooth} + Dose_{headphonesBluetooth}$$

Where:

- $Dose_{mobilecall}$  is the total dose from all mobile call activities
- $Dose_{phone}$  is the dose from the mobile phone (no Bluetooth)
- $Dose_{phoneBluetooth}$  is the dose from Bluetooth by the phone
- $Dose_{headphonesBluetooth}$  is the dose from Bluetooth by the Bluetooth headphones

**Value**

List with two values:

- "brain\_call\_dose" (brain RF-EMF dose from mobile calls in mJ/kg/day)
- "body\_call\_dose" (body RF-EMF dose from mobile calls in mJ/kg/day)

**See Also**

[get\\_mobilecall\\_phone\\_dose\(\)](#), [get\\_mobilecall\\_bt\\_phone\\_dose\(\)](#), [get\\_mobilecall\\_bt\\_headp\\_dose\(\)](#)

---

get\_mobilecall\_phone\_data\_pwr

*Calculate mobile phone output power from data mobile calling (no Bluetooth)*

---

**Description**

Calculates the output power of the mobile phone during data (3G, 4G, 5G) mobile calls in mW (No Bluetooth)

**Usage**

```
get_mobilecall_phone_data_pwr(urbanicity, use_5g, travel_time, params)
```

**Arguments**

|             |                                                                    |
|-------------|--------------------------------------------------------------------|
| urbanicity  | Urbanicity of home / workplace                                     |
| use_5g      | TRUE if participant uses 5G services on mobile phone, FALSE if not |
| travel_time | Time spent commuting in seconds per day                            |
| params      | Parameter list                                                     |

**Details**

The mobile phone output power during data mobile calling (no Bluetooth) is calculated as:

$$Power_{Data} = Prop_{3G} \times Power_{3G} + Prop_{4G} \times Power_{4G} + Prop_{5G} \times Power_{5G}$$

Where:

- $Prop_{3G}$ ,  $Prop_{4G}$ ,  $Prop_{5G}$  are the proportion of 3G, 4G and 5G during mobile calls, respectively
- $Power_{3G}$ ,  $Power_{4G}$ ,  $Power_{5G}$  are the output power of the mobile phone using 3G, 4G and 5G during mobile calls, respectively

**Value**

Mobile phone output power during data (3G, 4G, 5G) mobile calling (no Bluetooth) in mW



---

get\_mobilecall\_phone\_data\_sar

*Calculate SAR from data mobile calling (no Bluetooth)*


---

## Description

Calculates the tissue-specific absorption rate (SAR) for calling with a mobile phone (no Bluetooth) using mobile data (3G, 4G, 5G).

## Usage

```
get_mobilecall_phone_data_sar(
    ear_prop,
    headp_prop,
    speaker_prop,
    params,
    tissue_params
)
```

## Arguments

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| ear_prop      | Proportion of time mobile phone is held against ear during call                                         |
| headp_prop    | Proportion of time Bluetooth headphones are used during mobile call while phone is NOT held against ear |
| speaker_prop  | Proportion of time speaker mode is used during mobile call while phone is NOT held against ear          |
| params        | Parameter list                                                                                          |
| tissue_params | Tissue parameter list                                                                                   |

## Details

The SAR from mobile calling using data is calculated as:

$$SAR_{Data} = Prop_{Ear} \times SAR_{Ear} + Prop_{Speaker} \times SAR_{Speaker} + Prop_{Headphones} \times SAR_{Headphones}$$

Where:

- $Prop_{Ear}$ ,  $Prop_{Speaker}$ ,  $Prop_{Headphones}$  are the proportion of time phone is held against ear, used in speaker mode, and used with headphones, respectively
- $SAR_{Ear}$ ,  $SAR_{Speaker}$ ,  $SAR_{Headphones}$  are the SAR values from holding phone to ear, using it in speaker mode, and using it with headphones, respectively

## Value

SAR value from mobile phone during data (3G, 4G, 5G) calls in W/kg/W

---

get\_mobilecall\_phone\_dose

*Calculate RF-EMF dose from mobile phone (no Bluetooth)*


---

### Description

Calculates the total RF-EMF dose from mobile calling from the mobile phone (No Bluetooth)

### Usage

```
get_mobilecall_phone_dose(
    duration,
    ear_prop,
    speaker_prop,
    headp_prop,
    urbanicity,
    use_5g,
    travel_time,
    params,
    tissue_params
)
```

### Arguments

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| duration      | Duration of mobile phone call in seconds per day                                                        |
| ear_prop      | Proportion of time mobile phone is held against ear during call                                         |
| speaker_prop  | Proportion of time speaker mode is used during mobile call while phone is NOT held against ear          |
| headp_prop    | Proportion of time Bluetooth headphones are used during mobile call while phone is NOT held against ear |
| urbanicity    | Urbanicity of home / workplace                                                                          |
| use_5g        | TRUE if participant uses 5G services on mobile phone, FALSE if not                                      |
| travel_time   | Time spent commuting in seconds per day                                                                 |
| params        | Parameter list                                                                                          |
| tissue_params | Tissue-specific parameter list                                                                          |

### Details

The mobile call RF-EMF dose (no Bluetooth) is calculated as:

$$Dose = Duration \times SAR \times Power$$

Where:

- *Duration* is the duration of mobile phone call in seconds per day
- *SAR* is the specific absorption rate in W/kg/W
- *Power* is the output power of the mobile phone during mobile calling mW

**Value**

RF-EMF dose in mJ/kg/day

**See Also**

[get\\_mobilecall\\_phone\\_pwr\(\)](#), [get\\_mobilecall\\_phone\\_sar\(\)](#)

---

get\_mobilecall\_phone\_native\_pwr

*Calculate mobile phone output power from native mobile calling (no Bluetooth)*

---

**Description**

Calculates the output power of the mobile phone during native mobile calls in mW (No Bluetooth)

**Usage**

```
get_mobilecall_phone_native_pwr(urbanicity, travel_time, params)
```

**Arguments**

|             |                                         |
|-------------|-----------------------------------------|
| urbanicity  | Urbanicity of home / workplace          |
| travel_time | Time spent commuting in seconds per day |
| params      | Parameter list                          |

**Details**

The mobile phone output power during native mobile calling (no Bluetooth) is calculated as:

$$Power_{Data} = Prop_{2G} \times Power_{2G} + Prop_{3G} \times Power_{3G} + Prop_{4G} \times Power_{4G} + Prop_{5G} \times Power_{5G}$$

Where:

- $Prop_{2G}$ ,  $Prop_{3G}$ ,  $Prop_{4G}$ ,  $Prop_{5G}$  are the proportion of 2G, 3G, 4G and 5G during native mobile calls, respectively
- $Power_{2G}$ ,  $Power_{3G}$ ,  $Power_{4G}$ ,  $Power_{5G}$  are the output power of the mobile phone using 2G, 3G, 4G and 5G during native mobile calls, respectively

**Value**

Mobile phone output power during native mobile calling (no Bluetooth) in mW

---

get\_mobilecall\_phone\_native\_sar

*Calculate SAR from native mobile calling (no Bluetooth)*


---

## Description

Calculates the tissue-specific absorption rate (SAR) for calling with a mobile phone (no Bluetooth) using native network.

## Usage

```
get_mobilecall_phone_native_sar(
    ear_prop,
    headp_prop,
    speaker_prop,
    params,
    tissue_params
)
```

## Arguments

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| ear_prop      | Proportion of time mobile phone is held against ear during call                                         |
| headp_prop    | Proportion of time Bluetooth headphones are used during mobile call while phone is NOT held against ear |
| speaker_prop  | Proportion of time speaker mode is used during mobile call while phone is NOT held against ear          |
| params        | Parameter list                                                                                          |
| tissue_params | Tissue parameter list                                                                                   |

## Details

The SAR from native mobile calling is calculated as:

$$SAR_{Native} = Prop_{Ear} \times SAR_{Ear} + Prop_{Speaker} \times SAR_{Speaker} + Prop_{Headphones} \times SAR_{Headphones}$$

Where:

- $Prop_{Ear}$ ,  $Prop_{Speaker}$ ,  $Prop_{Headphones}$  are the proportion of time phone is held against ear, used in speaker mode, and used with headphones, respectively
- $SAR_{Ear}$ ,  $SAR_{Speaker}$ ,  $SAR_{Headphones}$  are the SAR values from holding phone to ear, using it in speaker mode, and using it with headphones, respectively

## Value

SAR value from mobile phone during native calls in W/kg/W

---

get\_mobilecall\_phone\_pwr

*Calculate output power from mobile phone during mobile call (no Bluetooth)*

---

## Description

Calculates the output power of the mobile phone during mobile calls in mW (No Bluetooth)

## Usage

```
get_mobilecall_phone_pwr(urbanicity, use_5g, travel_time, params)
```

## Arguments

|             |                                                                    |
|-------------|--------------------------------------------------------------------|
| urbanicity  | Urbanicity of home / workplace                                     |
| use_5g      | TRUE if participant uses 5G services on mobile phone, FALSE if not |
| travel_time | Time spent commuting in seconds per day                            |
| params      | Parameter list                                                     |

## Details

The mobile phone output power during mobile calling (no Bluetooth) is calculated as:

$$Power = Prop_{WiFi} \times Power_{WiFi} + Prop_{Data} \times Power_{Data} + Prop_{Native} \times Power_{Native}$$

Where:

- $Prop_{WiFi}$ ,  $Prop_{Data}$ ,  $Prop_{Native}$  are the proportion of time mobile calls are done using WiFi, mobile data, or native network, respectively.
- $Power_{WiFi}$ ,  $Power_{Data}$ ,  $Power_{Native}$  are the output power (mW) of the mobile phone using WiFi, mobile data, or native phone calls, respectively.

## Value

Mobile phone output power during mobile calling (no Bluetooth) in mW

## See Also

[get\\_mobilecall\\_phone\\_wifi\\_pwr\(\)](#), [get\\_mobilecall\\_phone\\_data\\_pwr\(\)](#), [get\\_mobilecall\\_phone\\_native\\_pwr\(\)](#)

---

get\_mobilecall\_phone\_sar

*Calculate SAR from mobile calling (no Bluetooth)*


---

### Description

Calculates the tissue-specific absorption rate (SAR) for calling with a mobile phone (no Bluetooth).

### Usage

```
get_mobilecall_phone_sar(
    ear_prop,
    headp_prop,
    speaker_prop,
    params,
    tissue_params
)
```

### Arguments

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| ear_prop      | Proportion of time mobile phone is held against ear during call                                         |
| headp_prop    | Proportion of time Bluetooth headphones are used during mobile call while phone is NOT held against ear |
| speaker_prop  | Proportion of time speaker mode is used during mobile call while phone is NOT held against ear          |
| params        | Parameter list                                                                                          |
| tissue_params | SAR values                                                                                              |

### Details

The SAR from mobile calling (no Bluetooth) is calculated as:

$$SAR = Prop_{WiFi} \times SAR_{WiFi} + Prop_{Data} \times SAR_{Data} + Prop_{Native} \times SAR_{Native}$$

Where:

- $Prop_{WiFi}$ ,  $Prop_{Data}$ ,  $Prop_{Native}$  are the proportion of WiFi, mobile data, and native network used during mobile calls, respectively
- $SAR_{WiFi}$ ,  $SAR_{Data}$ ,  $SAR_{Native}$  are the SAR values for WiFi, mobile data, and native network used during mobile calls, respectively

### Value

SAR from mobile calling (no Bluetooth) in W/kg/W

### See Also

[get\\_mobilecall\\_phone\\_wifi\\_sar\(\)](#), [get\\_mobilecall\\_phone\\_data\\_sar\(\)](#), [get\\_mobilecall\\_phone\\_native\\_sar\(\)](#)

---

get\_mobilecall\_phone\_wifi\_pwr

*Calculate mobile phone output power from WiFi mobile calling (no Bluetooth)*

---

## Description

Calculates the output power of the mobile phone during WiFi mobile calls in mW (No Bluetooth)

## Usage

```
get_mobilecall_phone_wifi_pwr(
    wifi_2_prop,
    wifi_5_prop,
    wifi_2_pwr,
    wifi_5_pwr,
    wifi_2_duty_cycle,
    wifi_5_duty_cycle
)
```

## Arguments

|                   |                           |
|-------------------|---------------------------|
| wifi_2_prop       | Proportion of 2.4GHz WiFi |
| wifi_5_prop       | Proportion of 5.0GHz WiFi |
| wifi_2_pwr        | 2.4GHz WiFi Call Power    |
| wifi_5_pwr        | 5.0GHz WiFi Call Power    |
| wifi_2_duty_cycle | 2.4GHz WiFi Duty Cycle    |
| wifi_5_duty_cycle | 5.0GHz WiFi Duty Cycle    |

## Details

The mobile phone output power during WiFi mobile calling (no Bluetooth) is calculated as:

$$Power_{WiFi} = Prop_{2.4GHz} \times Power_{2.4GHz} \times DutyCycle_{2.4GHz} + Prop_{5.0GHz} \times Power_{5.0GHz} \times DutyCycle_{5.0GHz}$$

Where:

- $Prop_{2.4GHz}$ ,  $Prop_{5.0GHz}$  are the proportion of 2.4GHz and 5.0GHz WiFi, respectively
- $Power_{2.4GHz}$ ,  $Power_{5.0GHz}$  are the output power of 2.4GHz and 5.0GHz WiFi calls from the mobile phone (no Bluetooth), respectively
- $DutyCycle_{2.4GHz}$ ,  $DutyCycle_{5.0GHz}$  are the duty cycle of 2.4GHz and 5.0GHz WiFi, respectively

## Value

Mobile phone output power during WiFi mobile calling (no Bluetooth) in mW

---

```
get_mobilecall_phone_wifi_sar
```

*Calculate SAR from WiFi mobile calling (no Bluetooth)*

---

### Description

Calculates the tissue-specific absorption rate (SAR) for calling with a mobile phone (no Bluetooth) using WiFi.

### Usage

```
get_mobilecall_phone_wifi_sar(
    ear_prop,
    headp_prop,
    speaker_prop,
    params,
    tissue_params
)
```

### Arguments

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| ear_prop      | Proportion of time mobile phone is held against ear during call                                         |
| headp_prop    | Proportion of time Bluetooth headphones are used during mobile call while phone is NOT held against ear |
| speaker_prop  | Proportion of time speaker mode is used during mobile call while phone is NOT held against ear          |
| params        | Parameter list                                                                                          |
| tissue_params | Tissue parameter list                                                                                   |

### Details

The SAR from mobile calling using WiFi is calculated as:

$$SAR_{WiFi} = Prop_{2.4GHz} \times SAR_{2.4GHz} + Prop_{5.0GHz} \times SAR_{5.0GHz}$$

Where:

- $Prop_{2.4GHz}$ ,  $Prop_{5.0GHz}$  are the proportion of 2.4GHz and 5.0GHz WiFi, respectively
- $SAR_{2.4GHz}$ ,  $SAR_{5.0GHz}$  are the SAR values from 2.4GHz and 5.0GHz WiFi calls from the mobile phone (no Bluetooth), respectively

### Value

SAR value from mobile phone during WiFi calls in W/kg/W



---

|                     |                                                                              |
|---------------------|------------------------------------------------------------------------------|
| get_mobiledata_dose | <i>Calculate dose from using data on mobile phone (mobile data and WiFi)</i> |
|---------------------|------------------------------------------------------------------------------|

---

### Description

Calculates the total RF-EMF dose from data transfer using a mobile phone. This includes mobile data (3G, 4G, 5G) and WiFi use. It does NOT include RF-EMF exposure from voice calling, which is calculated separately.

### Usage

```
get_mobiledata_dose(
    duration,
    use_5g,
    wifi_prop,
    urbanicity,
    act_pwr_props,
    travel_time,
    params
)
```

### Arguments

|               |                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------|
| duration      | Duration of mobile data use in seconds per day                                                   |
| use_5g        | TRUE if participant uses 5G services on mobile phone, FALSE if not                               |
| wifi_prop     | Proportion of time WiFi connection is used for data transfer (vs mobile data)                    |
| urbanicity    | Urbanicity of home / workplace                                                                   |
| act_pwr_props | List with proportion of time spent in low vs low-mid vs mid-high vs high output power activities |
| travel_time   | Time spent commuting in seconds per day                                                          |
| params        | Parameter list                                                                                   |

### Details

The mobile data RF-EMF dose is calculated as:

$$Dose_{mobiledata} = Duration_{mobiledata} \times Power_{mobiledata} \times SAR_{mobiledata}$$

Where:

- $Duration_{mobiledata}$  is the duration of mobile data use in seconds per day
- $Power_{mobiledata}$  is the output power of the mobile phone during mobile data use in mW
- $SAR_{mobiledata}$  is the specific absorption rate in W/kg/W

### Value

List with two values:

- "brain\_data\_dose" (brain RF-EMF dose from mobile data use in mJ/kg/day)
- "body\_data\_dose" (body RF-EMF dose from mobile data use in mJ/kg/day)

**See Also**

[get\\_mobiledata\\_pwr\(\)](#), [get\\_mobiledata\\_sar\(\)](#), [get\\_mobilecall\\_dose\(\)](#)

---

|                    |                                                      |
|--------------------|------------------------------------------------------|
| get_mobiledata_pwr | <i>Calculate mobile data aggregated output power</i> |
|--------------------|------------------------------------------------------|

---

**Description**

Calculates the output power of the mobile phone during mobile data use in mW

**Usage**

```
get_mobiledata_pwr(
  wifi_prop,
  use_5g,
  act_pwr_props,
  travel_time,
  urbanicity,
  params
)
```

**Arguments**

|               |                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------|
| wifi_prop     | Proportion of time WiFi connection is used for data transfer (vs mobile data)                    |
| use_5g        | TRUE if participant uses 5G services on mobile phone, FALSE if not                               |
| act_pwr_props | List with proportion of time spent in low vs low-mid vs mid-high vs high output power activities |
| travel_time   | Time spent commuting in seconds per day                                                          |
| urbanicity    | Urbanicity of home / workplace                                                                   |
| params        | parameter list                                                                                   |

**Details**

The mobile phone output power during mobile data use is calculated as:

$$Power = Prop_{WiFi} \times Power_{WiFi} + Prop_{Data} \times Power_{Data}$$

Where:

- $Prop_{WiFi}$ ,  $Prop_{Data}$  are the proportion of time mobile calls are done using WiFi and mobile data, respectively.
- $Power_{WiFi}$ ,  $Power_{Data}$  are the output power (mW) of the mobile phone using WiFi and mobile data respectively.

**Value**

Mobile phone output power during mobile data use in mW

**See Also**

[get\\_mpd\\_wifi\\_pwr\(\)](#), [get\\_mpd\\_data\\_pwr\(\)](#)

---

|                    |                                                                           |
|--------------------|---------------------------------------------------------------------------|
| get_mobiledata_sar | <i>Calculate SAR from data use on mobile phone (mobile data and WiFi)</i> |
|--------------------|---------------------------------------------------------------------------|

---

**Description**

Calculates specific absorption rate (SAR) from data use on mobile phone (mobile data and WiFi)

**Usage**

```
get_mobiledata_sar(wifi_prop, params, tissue_params)
```

**Arguments**

|               |                                                                               |
|---------------|-------------------------------------------------------------------------------|
| wifi_prop     | Proportion of time WiFi connection is used for data transfer (vs mobile data) |
| params        | Parameter list                                                                |
| tissue_params | Tissue-specific parameters (SAR-values)                                       |

**Details**

The SAR is calculated as follows:

$$SAR = Prop_{WiFi} \times SAR_{WiFi} + Prop_{Data} \times SAR_{data}$$

Where:  $Prop_{WiFi}$ ,  $Prop_{Data}$  are the WiFi and data use proportions  $SAR_{WiFi}$ ,  $SAR_{Data}$  are the SAR values from WiFi and from data

---

|                  |                                                                                         |
|------------------|-----------------------------------------------------------------------------------------|
| get_mpd_data_pwr | <i>Calculate mobile phone aggregated output power from mobile data (3G, 4G, 5G) use</i> |
|------------------|-----------------------------------------------------------------------------------------|

---

**Description**

Calculates mobile phone aggregated output power from mobile data (3G, 4G, 5G) use

**Usage**

```
get_mpd_data_pwr(use_5g, urbanicity, act_pwr_props, travel_time, params)
```

**Arguments**

|               |                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------|
| use_5g        | TRUE if participant uses 5G, FALSE otherwise                                                     |
| urbanicity    | Urbanicity of home/work environment                                                              |
| act_pwr_props | List with proportion of time spent in low vs low-mid vs mid-high vs high output power activities |
| travel_time   | Time spent commuting in seconds per day                                                          |
| params        | Device-specific parameter list                                                                   |

**Details**

The mobile phone output power from mobile data use is calculated as:

$$Power = Prop_{3G} \times Power_{3G} + Prop_{4G} \times Power_{4G} + Prop_{5G} \times Power_{5G}$$

Where:

- $Prop_{3G}$ ,  $Prop_{4G}$ ,  $Prop_{5G}$  are the proportions of 3G, 4G and 5G
- $Power_{3G}$ ,  $Power_{4G}$ ,  $Power_{5G}$  are the output powers of 3G, 4G and 5G (to be refined further)

**Value**

Output power from mobile data (3G, 4G, 5G) use on mobile phone in mW

---

|                  |                                                                     |
|------------------|---------------------------------------------------------------------|
| get_mpd_wifi_pwr | <i>Calculate mobile phone aggregated output power from WiFi use</i> |
|------------------|---------------------------------------------------------------------|

---

**Description**

Calculates mobile phone aggregated output power from WiFi use (2.4GHz, 5.0GHz)

**Usage**

```
get_mpd_wifi_pwr(act_pwr_props, params)
```

**Arguments**

|               |                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------|
| act_pwr_props | List with proportion of time spent in low vs low-mid vs mid-high vs high output power activities |
| params        | device-specific parameters                                                                       |

**Details**

The output power is calculated as:

$$Power_{WiFi} = Prop_{2.4GHz} \times Power_{2.4GHz} + Prop_{5.0GHz} \times Power_{5.0GHz}$$

Where:  $Prop_{2.4GHz}$ ,  $Prop_{5.0GHz}$  are the proportions of 2.4GHz vs 5.0GHz WiFi  $Power_{2.4GHz}$ ,  $Power_{5.0GHz}$  are the output power from 2.4GHz and 5.0GHz WiFi in mW, respectively

**Value**

Output power from WiFi use on mobile phone in mW

---

get\_other\_dose*Calculate Dose from other exposure sources*

---

**Description**

Calculate Dose from other exposure sources

**Usage**

```
get_other_dose(  
  duration_hotspot,  
  duration_smartwatch,  
  duration_tracker,  
  duration_vr,  
  duration_headphones,  
  duration_smarthome,  
  duration_gaming,  
  params  
)
```

**Arguments**

|                     |                                                |
|---------------------|------------------------------------------------|
| duration_hotspot    | daily hotspot use duration (s)                 |
| duration_smartwatch | daily smart watch use duration (s)             |
| duration_tracker    | daily tracker use duration (s)                 |
| duration_vr         | daily vr use duration (s)                      |
| duration_headphones | daily bluetooth headphone use duration (s)     |
| duration_smarthome  | daily time spend in smarthome (s)              |
| duration_gaming     | daily portable gaming console use duration (s) |
| params              | Parameter list                                 |

**Value**

List with brain dose and body dose in mJ/kg/day

---

|                    |                                                      |
|--------------------|------------------------------------------------------|
| get_smarthome_dose | <i>Calculate brain and body dose from smart home</i> |
|--------------------|------------------------------------------------------|

---

**Description**

Calculate brain and body dose from smart home

**Usage**

```
get_smarthome_dose(duration_smarthome, params)
```

**Arguments**

|                    |                                       |
|--------------------|---------------------------------------|
| duration_smarthome | Daily duration of being in smart home |
| params             | Parameter list                        |

**Value**

list with daily brain and body dose from smart home

---

|                   |                                               |
|-------------------|-----------------------------------------------|
| get_smarthome_pwr | <i>Calculate output power from smart home</i> |
|-------------------|-----------------------------------------------|

---

**Description**

Calculate output power from smart home

**Usage**

```
get_smarthome_pwr(params)
```

**Arguments**

|        |                |
|--------|----------------|
| params | Parameter list |
|--------|----------------|

**Value**

Aggregated output power smart home in mW

---

|                   |                                      |
|-------------------|--------------------------------------|
| get_smarthome_sar | <i>Calculate SAR from smart home</i> |
|-------------------|--------------------------------------|

---

**Description**

Calculate SAR from smart home

**Usage**

```
get_smarthome_sar(params, tissue_params)
```

**Arguments**

|               |                |
|---------------|----------------|
| params        | Parameter list |
| tissue_params | Tissue params  |

**Value**

tissue-specific SAR from smart home

---

|                     |                                                                                       |
|---------------------|---------------------------------------------------------------------------------------|
| get_smartwatch_dose | <i>Calculate brain and body dose from using smart watch connected to mobile phone</i> |
|---------------------|---------------------------------------------------------------------------------------|

---

**Description**

Important: exposure comes from both mobile phone and from smart watch.

**Usage**

```
get_smartwatch_dose(duration_smartwatch, params)
```

**Arguments**

|                     |                                                |
|---------------------|------------------------------------------------|
| duration_smartwatch | Daily duration of using smart watch in seconds |
| params              | Parameter list                                 |

**Value**

list with daily brain and body dose from smart watch use

---

`get_smartwatch_phone_dose`*Calculate contribution from mobile phone while using smart watch*

---

**Description**

Calculate contribution from mobile phone while using smart watch

**Usage**

```
get_smartwatch_phone_dose(duration_smartwatch, params, tissue_params)
```

**Arguments**

|                                  |                                             |
|----------------------------------|---------------------------------------------|
| <code>duration_smartwatch</code> | Daily duration of smartwatch use in seconds |
| <code>params</code>              | Parameter list                              |
| <code>tissue_params</code>       | tissue-specific parameter list              |

---

`get_smartwatch_phone_pwr`*Calculate power from smart watch (phone contribution only)*

---

**Description**

Calculate power from smart watch (phone contribution only)

**Usage**

```
get_smartwatch_phone_pwr(params)
```

**Arguments**

|                     |                |
|---------------------|----------------|
| <code>params</code> | Parameter list |
|---------------------|----------------|

**Value**

output power



---

`get_smartwatch_phone_sar`*Calculate SAR from smart watch (phone contribution only)*

---

**Description**

Calculate SAR from smart watch (phone contribution only)

**Usage**

```
get_smartwatch_phone_sar(params, tissue_params)
```

**Arguments**

`params`            Parameter list

`tissue_params`    Tissue-specific parameter list (SAR values)

**Value**

SAR value

---

`get_smartwatch_pwr`*Calculate power from smart watch (watch contribution only)*

---

**Description**

Calculate power from smart watch (watch contribution only)

**Usage**

```
get_smartwatch_pwr(params)
```

**Arguments**

`params`            Parameter list

**Value**

output power

---

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| get_smartwatch_sar | <i>Calculate SAR from smart watch (watch contribution only)</i> |
|--------------------|-----------------------------------------------------------------|

---

**Description**

Calculate SAR from smart watch (watch contribution only)

**Usage**

```
get_smartwatch_sar(params, tissue_params)
```

**Arguments**

|               |                                             |
|---------------|---------------------------------------------|
| params        | Parameter list                              |
| tissue_params | Tissue-specific parameter list (SAR values) |

**Value**

SAR value

---

|                           |                                                |
|---------------------------|------------------------------------------------|
| get_smartwatch_watch_dose | <i>Calculate contribution from smart watch</i> |
|---------------------------|------------------------------------------------|

---

**Description**

Calculate contribution from smart watch

**Usage**

```
get_smartwatch_watch_dose(duration_smartwatch, params, tissue_params)
```

**Arguments**

|                     |                                             |
|---------------------|---------------------------------------------|
| duration_smartwatch | Daily duration of smartwatch use in seconds |
| params              | Parameter list                              |
| tissue_params       | tissue-specific parameter list              |

---

|                 |                                       |
|-----------------|---------------------------------------|
| get_tablet_dose | <i>Calculate Dose from Tablet Use</i> |
|-----------------|---------------------------------------|

---

**Description**

Calculate Dose from Tablet Use

**Usage**

```
get_tablet_dose(duration, params)
```

**Arguments**

|          |                                   |
|----------|-----------------------------------|
| duration | Duration of tablet use in seconds |
| params   | Parameter list                    |

**Value**

List with brain dose and body dose in mJ/kg/day

---

|                |                                          |
|----------------|------------------------------------------|
| get_tablet_pwr | <i>Calculate Tablet Aggregated Power</i> |
|----------------|------------------------------------------|

---

**Description**

Calculate Tablet Aggregated Power

**Usage**

```
get_tablet_pwr(params)
```

**Arguments**

|        |       |
|--------|-------|
| params | descr |
|--------|-------|

**Value**

Aggregated tablet power

---

|                |                                        |
|----------------|----------------------------------------|
| get_tablet_sar | <i>Calculate Tablet Aggregated SAR</i> |
|----------------|----------------------------------------|

---

**Description**

Calculate Tablet Aggregated SAR

**Usage**

```
get_tablet_sar(params, tissue_params)
```

**Arguments**

|               |       |
|---------------|-------|
| params        | descr |
| tissue_params | descr |

**Value**

Aggregated tablet SAR

---

|                |                                                                                             |
|----------------|---------------------------------------------------------------------------------------------|
| get_total_dose | <i>Calculate Total RF-EMF Dose for Brain and Body from All Sources For a SINGLE SAMPLE!</i> |
|----------------|---------------------------------------------------------------------------------------------|

---

**Description**

Calculate Total RF-EMF Dose for Brain and Body from All Sources For a SINGLE SAMPLE!

**Usage**

```
get_total_dose(sample, params)
```

**Arguments**

|        |                                              |
|--------|----------------------------------------------|
| sample | A list with input values for a single sample |
| params | A parameter list                             |

**Value**

A list with results for brain and body dose for a single sample

---

|                  |                                                                                            |
|------------------|--------------------------------------------------------------------------------------------|
| get_tracker_dose | <i>Calculate brain and body dose from using personal tracker connected to mobile phone</i> |
|------------------|--------------------------------------------------------------------------------------------|

---

**Description**

Important: exposure comes from both mobile phone and from smart watch.

**Usage**

```
get_tracker_dose(duration_tracker, params)
```

**Arguments**

|                  |                                            |
|------------------|--------------------------------------------|
| duration_tracker | Daily duration of using tracker in seconds |
| params           | Parameter list                             |

**Value**

list with daily brain and body dose from tracker use

---

|                        |                                                                     |
|------------------------|---------------------------------------------------------------------|
| get_tracker_phone_dose | <i>Calculate contribution from mobile phone while using tracker</i> |
|------------------------|---------------------------------------------------------------------|

---

**Description**

Calculate contribution from mobile phone while using tracker

**Usage**

```
get_tracker_phone_dose(duration_tracker, params, tissue_params)
```

**Arguments**

|                  |                                          |
|------------------|------------------------------------------|
| duration_tracker | Daily duration of tracker use in seconds |
| params           | Parameter list                           |
| tissue_params    | tissue-specific parameter list           |

---

|                       |                                                               |
|-----------------------|---------------------------------------------------------------|
| get_tracker_phone_pwr | <i>Calculate power from tracker (phone contribution only)</i> |
|-----------------------|---------------------------------------------------------------|

---

**Description**

Calculate power from tracker (phone contribution only)

**Usage**

get\_tracker\_phone\_pwr(params)

**Arguments**

|        |                |
|--------|----------------|
| params | Parameter list |
|--------|----------------|

**Value**

output power

---

|                       |                                                             |
|-----------------------|-------------------------------------------------------------|
| get_tracker_phone_sar | <i>Calculate SAR from tracker (phone contribution only)</i> |
|-----------------------|-------------------------------------------------------------|

---

**Description**

Calculate SAR from tracker (phone contribution only)

**Usage**

get\_tracker\_phone\_sar(params, tissue\_params)

**Arguments**

|               |                                             |
|---------------|---------------------------------------------|
| params        | Parameter list                              |
| tissue_params | Tissue-specific parameter list (SAR values) |

**Value**

SAR value

---

|                 |                                                                 |
|-----------------|-----------------------------------------------------------------|
| get_tracker_pwr | <i>Calculate power from tracker (tracker contribution only)</i> |
|-----------------|-----------------------------------------------------------------|

---

**Description**

Calculate power from tracker (tracker contribution only)

**Usage**

```
get_tracker_pwr(params)
```

**Arguments**

|        |                |
|--------|----------------|
| params | Parameter list |
|--------|----------------|

**Value**

output power

---

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| get_tracker_sar | <i>Calculate SAR from tracker (watch contribution only)</i> |
|-----------------|-------------------------------------------------------------|

---

**Description**

Calculate SAR from tracker (watch contribution only)

**Usage**

```
get_tracker_sar(params, tissue_params)
```

**Arguments**

|               |                                             |
|---------------|---------------------------------------------|
| params        | Parameter list                              |
| tissue_params | Tissue-specific parameter list (SAR values) |

**Value**

SAR value

---

`get_tracker_tracker_dose`*Calculate contribution from tracker*

---

**Description**

Calculate contribution from tracker

**Usage**

```
get_tracker_tracker_dose(duration_tracker, params, tissue_params)
```

**Arguments**

|                               |                                          |
|-------------------------------|------------------------------------------|
| <code>duration_tracker</code> | Daily duration of tracker use in seconds |
| <code>params</code>           | Parameter list                           |
| <code>tissue_params</code>    | tissue-specific parameter list           |

---

`get_vr_dose`*Calculate brain and body dose from using VR headset*

---

**Description**

Calculate brain and body dose from using VR headset

**Usage**

```
get_vr_dose(duration_vr, params)
```

**Arguments**

|                          |                                               |
|--------------------------|-----------------------------------------------|
| <code>duration_vr</code> | Daily duration of using VR headset in seconds |
| <code>params</code>      | Parameter list                                |

**Value**

list with daily brain and body dose from VR headset



---

`get_vr_pwr`*Calculate output power from virtual reality headset*

---

**Description**

Calculate output power from virtual reality headset

**Usage**

```
get_vr_pwr(params)
```

**Arguments**

|                     |                |
|---------------------|----------------|
| <code>params</code> | Parameter list |
|---------------------|----------------|

**Value**

Aggregated output power of VR headset in mW

---

`get_vr_sar`*Calculate SAR from VR headset*

---

**Description**

Calculate SAR from VR headset

**Usage**

```
get_vr_sar(params, tissue_params)
```

**Arguments**

|                            |                |
|----------------------------|----------------|
| <code>params</code>        | Parameter list |
| <code>tissue_params</code> | Tissue params  |

**Value**

tissue-specific SAR from VR headset

---

|               |                                     |
|---------------|-------------------------------------|
| get_wifi_dose | <i>Calculate Dose from wifi Use</i> |
|---------------|-------------------------------------|

---

**Description**

Calculate Dose from wifi Use

**Usage**

```
get_wifi_dose(travel_time, params)
```

**Arguments**

|             |                                |
|-------------|--------------------------------|
| travel_time | Daily time spent commuting (s) |
| params      | Parameter list                 |

**Value**

List with brain dose and body dose in mJ/kg/day

---

|              |                                        |
|--------------|----------------------------------------|
| get_wifi_pwr | <i>Calculate WiFi Aggregated Power</i> |
|--------------|----------------------------------------|

---

**Description**

Calculate WiFi Aggregated Power

**Usage**

```
get_wifi_pwr(params)
```

**Arguments**

|        |                |
|--------|----------------|
| params | parameter list |
|--------|----------------|

**Value**

aggregated output power wifi

---

|              |                                      |
|--------------|--------------------------------------|
| get_wifi_sar | <i>Calculate WiFi Aggregated SAR</i> |
|--------------|--------------------------------------|

---

**Description**

Calculate WiFi Aggregated SAR

**Usage**

```
get_wifi_sar(params, tissue_params)
```

**Arguments**

|               |             |
|---------------|-------------|
| params        | description |
| tissue_params | description |

**Value**

description

---

|                    |                                        |
|--------------------|----------------------------------------|
| load_device_params | <i>Load device-specific parameters</i> |
|--------------------|----------------------------------------|

---

**Description**

Load device-specific parameters

**Usage**

```
load_device_params(params, device_type)
```

**Arguments**

|             |                |
|-------------|----------------|
| params      | Parameter list |
| device_type | Device type    |

**Value**

Parameter list specific to selected device

---

|             |                           |
|-------------|---------------------------|
| load_params | <i>Loading parameters</i> |
|-------------|---------------------------|

---

**Description**

Loading parameters

**Usage**

```
load_params(filename)
```

**Arguments**

|          |                                                         |
|----------|---------------------------------------------------------|
| filename | Name of parameter file to load (must be in yaml format) |
|----------|---------------------------------------------------------|

**Value**

List of parameters loaded from yaml input file

---

|                    |                                        |
|--------------------|----------------------------------------|
| load_tissue_params | <i>Load tissue-specific parameters</i> |
|--------------------|----------------------------------------|

---

**Description**

Load tissue-specific parameters

**Usage**

```
load_tissue_params(params, device_type, tissue_name)
```

**Arguments**

|             |                                |
|-------------|--------------------------------|
| params      | Device-specific parameter list |
| device_type | Name of device                 |
| tissue_name | Name of tissue                 |

**Value**

Parameter list specific to selected device and tissue

---

|                   |                                                    |
|-------------------|----------------------------------------------------|
| recode_urbanicity | <i>Check and re-code input values - urbanicity</i> |
|-------------------|----------------------------------------------------|

---

**Description**

Check and re-code input values - urbanicity

**Usage**

```
recode_urbanicity(urbanicity)
```

**Arguments**

|            |       |
|------------|-------|
| urbanicity | descr |
|------------|-------|

**Value**

re-coded urbanicity (binary variables)

---

|                          |                                    |
|--------------------------|------------------------------------|
| save_default_params_file | <i>Load default parameter list</i> |
|--------------------------|------------------------------------|

---

**Description**

Load default parameter list

**Usage**

```
save_default_params_file(dest_file = NULL)
```

**Arguments**

|           |                                                     |
|-----------|-----------------------------------------------------|
| dest_file | location where downloaded YAML file should be saved |
|-----------|-----------------------------------------------------|

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